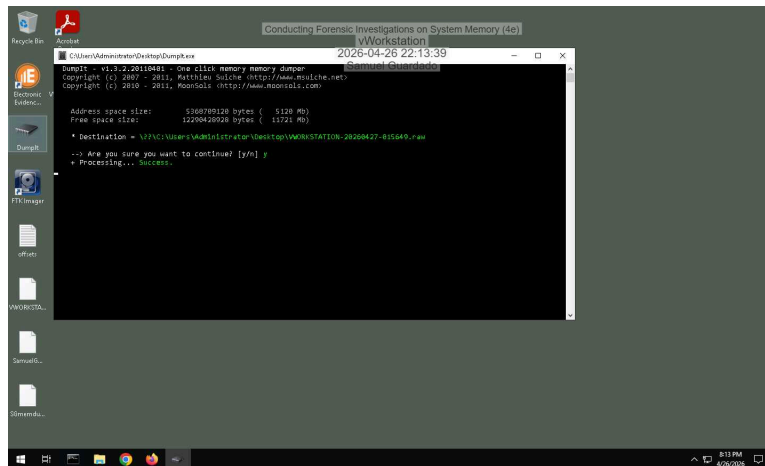


Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

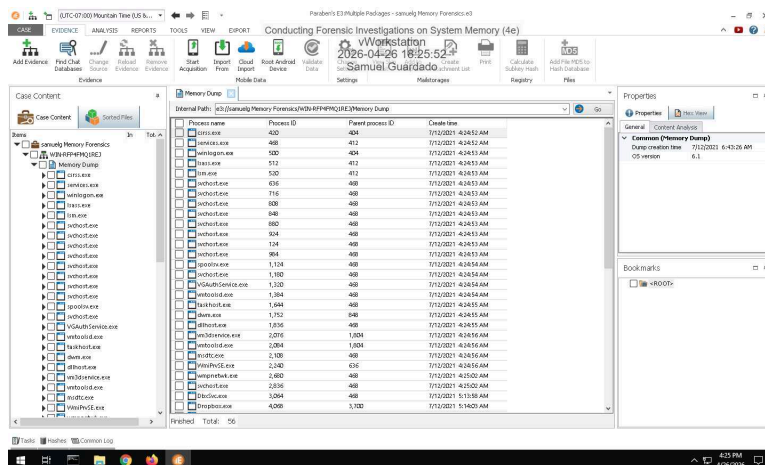
Time on Task: 3 hours, 18 minutes
Progress: 100%

Part 1: Capture Memory using DumpIt

3. Make a screen capture showing the **DumpIt** success notification.



8. **Make a screen capture** showing the **list of processes in the memory dump**.

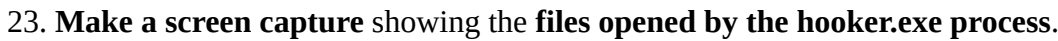


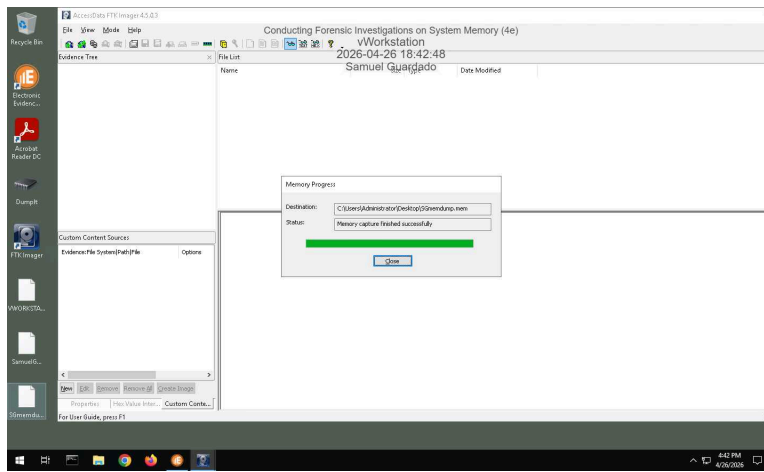
10. **Record** the start times for the oldest process and the newest process.

Oldest process was on 7/12/2021 at 4:24:49 AM, and the latest was, the latest process started on 7/12/2021 6:42:43 AM.

15. **Document** your findings for the conhost.exe process. What is it and what is it used for?

21. **Make a screen capture** showing the **registry keys** opened by the **Hooker.exe** process.





Part 2: Analyze Memory using Volatility

7. **Document** your findings for the rvkl.exe process. What is it and what is it used for?

rvkl.exe is a keylogger/monitoring tool that can record your keystrokes, and might send the information to a remote user in an e-mail.

9. **Document** whether any processes are flagged as hidden.

Services.exe, Isass.exe, Ism.exe are flagged as hidden.

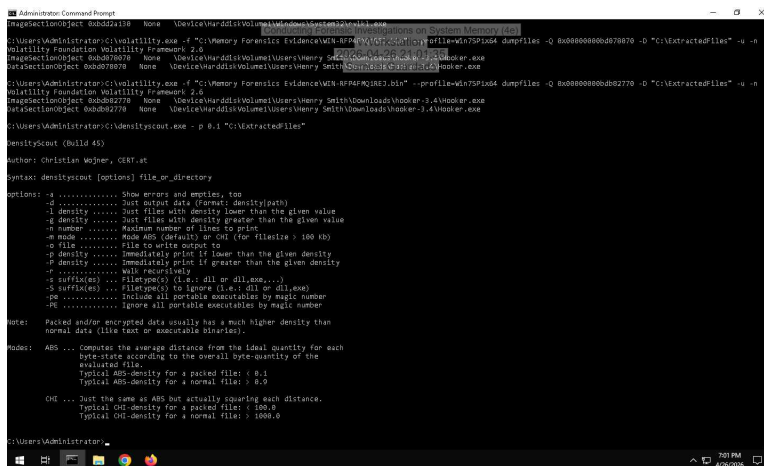
12. **Document** whether the netscan module displays network usage associated with the Hooker.exe or rvkl.exe processes.

The netscan module displays network usage associated with the Hooker.exe or rvkl.exe processes through Chrome and Dropbox.

15. **Document** any information you were able to gather about port 56610.

is a high-numbered, dynamic, or private port typically used for temporary connections rather than well known services. It is usually unregistered, but it is associated with Apple's Xsan Filesystem access.

26. **Make a screen capture** showing the DensityScout results.



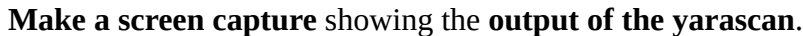
Section 3: Challenge and Analysis

Part 1: Identify Malicious Connections

the three processes that connected to 205.134.253.10:4444 where fixtureCompute, QaNoQBC.exe, and dlllhost.exe.

Port 4444 is usually used by Metasploit, a security software programming tool used for identifying vulnerabilities and help with penetration testing.

Make a screen capture showing the `fixtureComputer.exe` process, and all those below it, in the `pslist` output.



4

0544 taskeng.exe	28 SeImpersonatePrivilege	Present,Enabled,Default	Impersonate a client after authentication
0544 taskeng.exe	38 SecCreateLocalPrivilege	Present,Enabled,Default	Create global objects
0544 taskeng.exe	11 SeTrusteeControlAccessPrivilege	Present,Enabled,Default	Access Credential Manager as a trusted caller
0544 taskeng.exe	32 SeLabelPrivilege	Present	Modify the mandatory integrity level of an object
0544 taskeng.exe	13 SeIncreaseBasePriorityPrivilege	Present,Enabled,Default	Allocate more memory for user applications
0544 taskeng.exe	34 SeTimeZonePrivilege	Present,Enabled,Default	Adjust the time zone of the computer's internal clock
0544 taskeng.exe	15 SeCreateSymbolicLinkPrivilege	Present,Enabled,Default	Required to create a symbolic link
0772 Hooker.exe	2 SecCreateObjectPrivilege	Present	Create a token object
0772 Hooker.exe	3 SeDebugPrivilege	Present	Replace a process-level token
0772 Hooker.exe	4 SeLockMemoryPrivilege	Present	Lock pages in memory
0772 Hooker.exe	5 SeIncreaseQuotaPrivilege	Present	Increase quotas
0772 Hooker.exe	6 SeBackupPrivilege	Present	Add write permissions to the domain
0772 Hooker.exe	7 SeRemoteAdminPrivilege	Present	Act as part of the operating system
0772 Hooker.exe	8 SeSecurityPrivilege	Present	Manage auditing and security log
0772 Hooker.exe	9 SeFileOwnerPrivilege	Present	Take ownership of files/objects
0772 Hooker.exe	10 SeDevicePrivilege	Present	Load and unload service drivers
0772 Hooker.exe	11 SeSystemPrivilege	Present	Profile system performance
0772 Hooker.exe	12 SeSystemTimePrivilege	Present	Change the system time
0772 Hooker.exe	13 SeProfileSingleProcessPrivilege	Present	Profile a single process
0772 Hooker.exe	14 SeIncreaseBasePriorityPrivilege	Present	Increase scheduling priority
0772 Hooker.exe	15 SecCreatePageFilePrivilege	Present	Create a pagefile
0772 Hooker.exe	16 SeCreateNamedPipePrivilege	Present	Create permanent shared objects
0772 Hooker.exe	17 SeBackupPrivilege	Present	Backup files and directories
0772 Hooker.exe	18 SeRestorePrivilege	Present	Restore files and directories
0772 Hooker.exe	19 SeShutdownPrivilege	Present	Shut down the system
0772 Hooker.exe	20 SeDebugPrivilege	Present	Debug program
0772 Hooker.exe	21 SeAuditPrivilege	Present	Generate security audits
0772 Hooker.exe	22 SeSystemEnvironmentPrivilege	Present	Edit firmware environment values
0772 Hooker.exe	23 SeChangeEventPrivilege	Present,Enabled,Default	Receive notifications of changes to files or directories
0772 Hooker.exe	24 SeRemoteShutdownPrivilege	Present	Force shutdown from a remote system
0772 Hooker.exe	25 SeRemoteAdminPrivilege	Present	Remove computer from docking station
0772 Hooker.exe	26 SeSyncAgentPrivilege	Present	Synch directory service data
0772 Hooker.exe	27 SeTrustedCredManAccessPrivilege	Present	Enable user accounts to be trusted for delegation
0772 Hooker.exe	28 SeManageVolumePrivilege	Present	Manage the files on a volume
0772 Hooker.exe	29 SeImpersonatePrivilege	Present,Enabled,Default	Impersonate a client after authentication
0772 Hooker.exe	38 SecCreateLocalPrivilege	Present,Enabled,Default	Create global objects
0772 Hooker.exe	11 SeTrusteeControlAccessPrivilege	Present	Access Credential Manager as a trusted caller
0772 Hooker.exe	32 SeLabelPrivilege	Present	Modify the mandatory integrity level of an object
0772 Hooker.exe	13 SeIncreaseBasePriorityPrivilege	Present	Allocate more memory for user applications
0772 Hooker.exe	34 SeTimeZonePrivilege	Present	Adjust the time zone of the computer's internal clock
0772 Hooker.exe	15 SeCreateSymbolicLinkPrivilege	Present	Required to create a symbolic link
2608 cmd.exe	2 SecCreateObjectPrivilege	Present	Create a token object
2608 cmd.exe	3 SeDebugPrivilege	Present	Replace a process-level token
2608 cmd.exe	4 SeLockMemoryPrivilege	Present	Lock pages in memory
2608 cmd.exe	5 SeIncreaseQuotaPrivilege	Present	Increase quotas
2608 cmd.exe	6 SeBackupPrivilege	Present	Add write permissions to the domain
2608 cmd.exe	7 SeRemoteAdminPrivilege	Present	Act as part of the operating system
2608 cmd.exe	8 SeSecurityPrivilege	Present	Manage auditing and security log
2608 cmd.exe	9 SeFileOwnerPrivilege	Present	Take ownership of files/objects
2608 cmd.exe	10 SeDevicePrivilege	Present	Load and unload service drivers